From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Main), 2014

FULL TEST – V

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

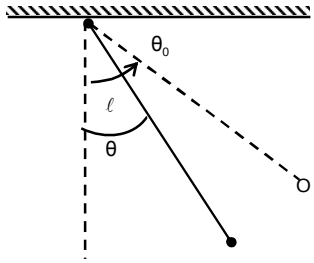
SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

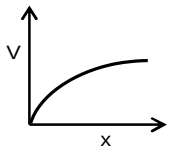
1. The orbital radius of a hydrogen atom is 0.847 nm. The electron makes a transition from this state to a lower state. How many possible wavelengths are emitted?
(A) 1 (B) 2
(C) 3 (D) 4

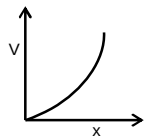
2. A point mass is attached to an inextensible string of length ℓ. The particle is released from a position where the string is initially horizontal. At some time string makes an angle θ with vertical & let u be the speed of particle at that instant then

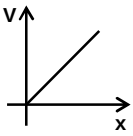


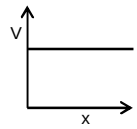
(A) $T\cos\theta = mg$ (T is tension in spring)
(B) $mg\cos\theta = T$
(C) T = 3mg cos θ
(D) T = 2mg cos θ

3. A particle starts from origin at t=0 according to law $x = A(1-e^{-\lambda t})$ graph between its velocity and displacement is

(A) 

(B) 

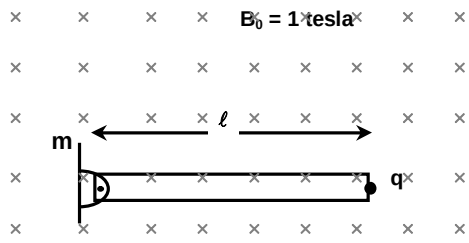
(C) 

(D) 

4. A constant and uniform magnetic field of strength, B = 1T is established in a cylindrical region of radius 10 cm. A rectangular conducting wire loop of size 30 cm $\times$ 30 cm is taken and moved so that all the magnetic field lines pass through this loop once. The resistance of the loop is 200 Ω. The net charge flowing through the loop is
(A) 4.5 C (B) 4.5×10^{-4}C
(C) $\frac{\pi}{2}$C (D) $\frac{\pi}{2}\times 10^{-4}$C

Rough Work

5. A uniform non conducting bar of mass 1 kg, length 1 m is hinged at one end so that it can rotate freely in vertical plane. A charge $\sqrt{\frac{3}{10}}$C is placed on the other end. If the bar is released from horizontal position the force exerted on the hinge, when the rod becomes vertical, is



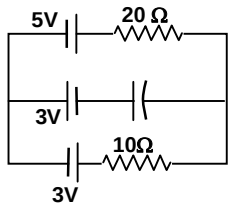
(A) $\frac{\mu_0 q}{4\pi\ell}\sqrt{\frac{3g}{\ell}}$ (B) $\frac{\mu_0 q}{2\pi\ell}\sqrt{\frac{3g}{\ell}}$

(C) $\frac{\mu_0 q}{4\pi}\sqrt{3g\ell}$ (D) $\frac{\mu_0 q}{2\pi}\sqrt{3g\ell}$

6. The capacitor is the ratio uncharged of the current through the 10 Ω resistance through the 20 Ω resistance is, initially,

(A) 3/2 (B) 6/5

(C) 1/1 (D) 2/3



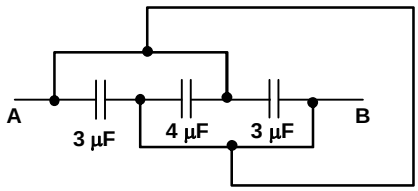
7. The equivalent capacitance of the circuit between A and B is

(A) 10 μF

(B) 5 μF

(C) 15 μF

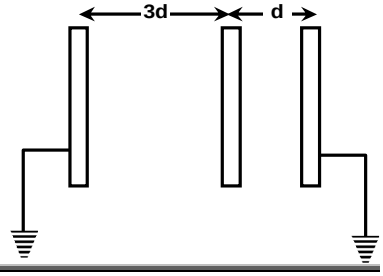
(D) 20 μF



8. Three similar plates are arranged as shown in the figure. If area of each plate is A and a charge +Q is given to middle plate then potential of middle plate is

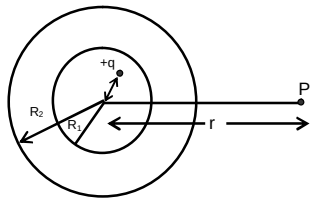
(A) $\frac{3}{4}\frac{Qd}{A\in_0}$ (B) $\frac{Qd}{4A\in_0}$

(C) $\frac{Qd}{3A\in_0}$ (D) $\frac{Qd}{2A\in_0}$



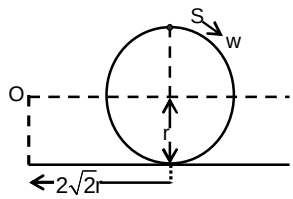
Rough Work

9. A conducting shell having inner radius R_1 and outer radius R_2 contains a charge + q which is placed at a distance x from its centre. Field at an exterior point P which is situated at a distance r from centre of shell $(r > R_2)$



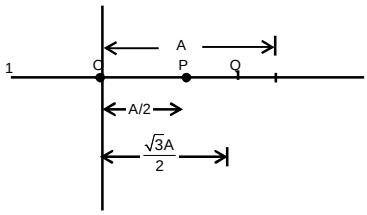
(A) depends on x
(B) depends on R_1 and R_2
(C) depends only or r
(D) None.

10. Consider the situation shown in the figure. The source S emits a wave pulse having frequency f_0 when it is at highest point on the ring (which is performing pure rolling). The frequency of that pulse heard by an observer at O is



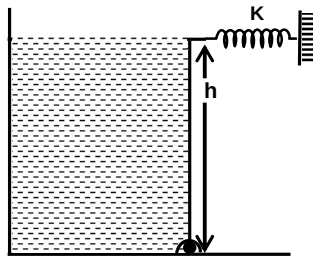
(A) $cfo/\left(3c+2\sqrt{2}c\right)$
(B) $3cfo/\left(3c+2\sqrt{2}c\right)$
(C) $cfo/\left(3c-2\sqrt{2}c\right)$
(D) $3fo/\left(3c-2\sqrt{2}c\right)$

11. A particle is performing simple harmonic motion with amplitude A and time period T. Time taken by particle in going from point P to Q is



(A) T/3 (B) T/6
(C) T/12 (D) T/18

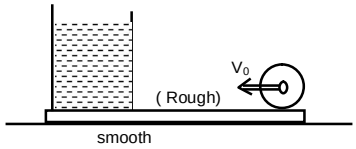
12. A plank of length h and width b (perpendicular to plane of paper is hinged at one end. The upper end is attached with a spring of spring constant k and water is filled up to high h. If plank is in equilibrium in vertical position then potential energy stored in spring is (ignore effect of atmospheric pressure)



(A) $\frac{b^2\rho^2g^2h^4}{18}$ (B) $\frac{b^2\rho^2g^2h^4}{36k}$
(C) $\frac{b^2\rho^2g^2h^4}{72k}$ (D) $\frac{b^2\rho^2g^2h^4}{128k}$

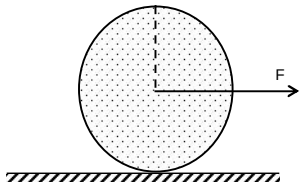
Rough Work

13. A beaker containing liquid is fixed on a long plank which is placed on a smooth ground. A disk is given a velocity V_0 on upper rough surface of plank. When pure rolling starts the surface of liquid is



(A) horizontal
(B) curved
(C) inclined
(D) None of the above

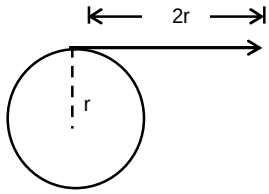
14. A uniform disk of mass m is resting on a rough horizontal surface. A force F is applied at the centre of disc at t=0. If disc performs pure rolling then power developed by F at any time t is



(A) $\frac{3F^2t}{2m}$ (B) $\frac{2F^2t}{3m}$

(C) $\frac{4F^2t}{3m}$ (D) $\frac{aF^2t}{43m}$

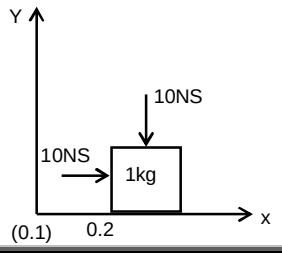
15. A uniform bar of mass m and length 2r is attached to ring of mass m and radius r. such that it is tangential to the ring and both are in same plane. Moment of inertia of this assembly about an axis passing through centre of ring and perpendicular to plane is -



(A) $\frac{10}{3}mr^2$ (B) $\frac{7}{3}mr^2$

(C) $\frac{11}{3}mr^2$ (D) $\frac{19}{3}mr^2$

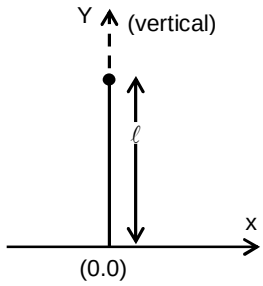
16. A block of mass 1kg is resting on a rough horizontal surface with coefficient of friction 0.2. If an impulse $\left(10\hat{i}-10\hat{j}\right)$ N - s is given to block then velocity of block just after the given impulse is



(A) 10 m/sec. (B) 0 m/sec.
(C) 8 m/sec. (D) 4m/sec.

Rough Work

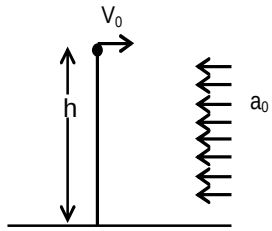
17. A mass m is attached to one end of a light rod. This assembly is placed on the origin as shown in the figure. If all surfaces are smooth then the position of lower end when the rod becomes horizontal after its upper end has been disturbed gently towards right is



(A) $\ell,0$ (B) $\frac{\ell}{2},0$

(C) $-\ell,0$ (D) $-\frac{\ell}{2},0$

18. A particle is projected horizontally from a tower with velocity V_0. Wind is blowing in opposite direction and is providing a constant horizontal acceleration a_0. If particle strikes the ground vertically then height of building is



(A) $V_0^2 g / a_0$ (B) $\frac{V_0^2 g}{2a_0}$

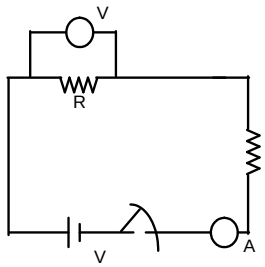
(C) $\frac{2V_0^2 g}{a_0}$ (D) $\frac{V_0^2}{g}$

19. Two waves given by equations $y_1 = 3\sin(100t - 20x)m$ and $y_2 = 4\cos(100t - 20x)m$ interfere at a point then amplitude of the resultant wave is

(A) 7m (B) 1m

(C) 25m (D) 5m.

20. The potential deference across the resistor as measured by a voltmeter is $60 \pm 1.5V$ and the current through the resistor as measured by an ammeter is (2 ± 0.01) A. what is the maximum % error in the measured value of R?



(A) 3% (B) 1.5%

(C) 2.5% (D) None of these

21. Which of the following is equal to the minimum kinetic energy needed by a hydrogen atom in ground state to produce a photon after collision with another identical hydrogen atom at rest?

(A) 6.8eV (B) 3.4eV

(C) 20.4eV (D) 2.04eV

Rough Work

22. Suppose a parent nucleus decays into a daughter nucleus with a decay constant of λ_p and the daughter nucleus of λ_d. At any instant the ratio of parent to daughter nuclei is observed to be 2. What should be the ratio λ_p / λ_d if the number, of daughter nucleus is constant?

(A) 2 (B) $\frac{1}{2}$

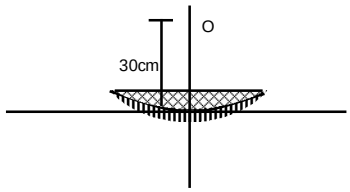
(C) $\frac{1}{4}$ (D) 4.

23. The half life of a radioactive substance A is 2 hour and that of B 4 hour. What fraction of the original activity of an equi-molar mixture of A and B remains after 12 hour?

(A) $\frac{9}{64}$ (B) $\frac{5}{96}$

(C) $\frac{1}{16}$ (D) $\frac{1}{16\sqrt{2}}$

24. A concave mirror is placed in a medium of refractive index $\mu_1 = 1.2$. An object placed at 30 cm from pole of the mirror and is image coincide. Now a liquid of refractive index μ is placed in the concave mirror in the shaded region and it is found that object needs to be shifted by 7.5 cm below O for the object and image to coincide. The refractive index of the liquid μ is



(A) $\frac{4}{3}$ (B) 1.2

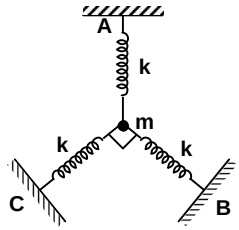
(C) 1.6 (D) None of the above.

25. An object is placed at 30 cm from a convex lens of focal length 15cm. On the other side of the lens a convex mirror of focal length 12cm is placed so that the principal axis of both coincide. It is observed that the object and image coincide. What is the separation between the lens and mirror?

(A) 6cm (B) 30cm

(C) both (A) and (B) (D) none of the above

26. A particle of mass m is attached to three identical springs of spring constant k connected to the three vertices of an equilateral triangle ABC. The mass m is pressed against block C and released. The time period of small oscillations is



(A) $2\pi\sqrt{\frac{3m}{k}}$ (B) $2\pi\sqrt{\frac{m}{3k}}$

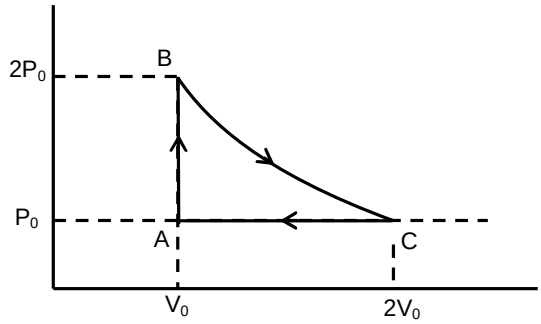
(C) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{2m}{3k}}$

Rough Work

27. A tuning fork of frequency 330 Hz is observed to resonate with an open organ pipe. The length of the open pipe could be (speed of sound in air = 330 m/s)
(A) 1/2 m (B) 3/2 m
(C) 3/4 m (D) 1 m

28. An inductor of inductance $\frac{1}{\pi^2}$mH is connected in series with a capacitor of 2.5 μF charged to 30 V and the circuit is completed (t = 0). The time when the voltage across the capacitor becomes 15 V for the first time is
(A) $\frac{1}{12}\times10^{-4}$s (B) $\frac{1}{2}\times10^{-4}$s
(C) $\frac{1}{6}\times10^{-4}$s (D) $\frac{1}{8}\times10^{-4}$s

29. 1 mole of a mono atomic ideal gas undergoes a cyclic process shown in the diagram. The process BC is isothermal and the temperature at B is 800 K. The work done by the gas during the process ABC equals
(A) 1600 Rln2
(B) 3200 Rln2
(C) 800 Rln2
(D) zero



30. A calorimeter of water equivalent 20g is filled with 230 g water at 20°C. An unknown solid of mass 150g heated to a temperature of 75°C is put into the calorimeter and the system is thermally insulated from the surroundings. The final temperature of the system is found to be 25°C. The water equivalent of the solid is
(A) 150 g (B) 75g
(C) 45 g (D) 25 g

Rough Work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice** questions numbered **1 to 30.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Which of the following is resistant to hydrolysis at room temperature?
   (A) SF_6 (B) $SiCl_4$
   (C) PCl_5 (D) BCl_3

2. The complex ion which exhibits geometrical isomerism as well as optical isomerism is:
   (A) $[Cr(NH_3)_4Cl_2]^+$ (B) $[Pt(NH_3)_2Cl_2]$
   (C) $[Pt(NH_3)_2(Br_2)_2Cl_2]$ (D) $[Ca(EDTA)]^{2-}$

3. Select the reaction (s) which would liberate hydrogen gas?
   (I) $SiH_4 + H_2O \longrightarrow ?$
   (II) $Al + NaOH \longrightarrow ?$
   (III) $Zn + H_2SO_4\,(\text{dilute}) \longrightarrow ?$
   (IV) $XeF_6 + H_2O \longrightarrow ?$
   (A) II, III (B) I, II, III
   (C) I, IV (D) I, II, III, IV

4. The metal which is extracted by hydrometallurgy as well as pyrometallurgy is:
   (A) Mg (B) Fe
   (C) Pb (D) Cu

5. Which of the given set of compounds does not exist?
   (A) H_2S, $PbCl_4$, FeI_3 (B) HgI_2, FeI_3, PbI_4
   (C) PH_5, SCl_6, $BiCl_5$ (D) BiI_3, Al_2Br_6, Fe_2Cl_6

Rough Work

6. The correct order of increasing bond length is:

(A) $H_2 > H_2^+ > H_2^-$ (B) $H_2 < H_2^+ < H_2^-$

(C) $H_2 < H_2^+ = H_2^-$ (D) $H_2 < H_2^- < H_2^+$

7. $B_2H_6 + \underset{(\text{Lewis base})}{2L:} \longrightarrow 2BH_3L$

L : would be

(A) NH_3 (B) CH_3NH_2

(C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

8. Which of the following reaction is exothermic in nature?

(A) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

(B) $2Al + Fe_2O_3 \xrightarrow{\Delta} 2Fe + Al_2O_3$

(C) $2Na + 2H_2O \longrightarrow 2NaOH + H_2 \uparrow$

(D) All of these

9. Which of the following set of cations can not be precipitated by treatment with sodium thiosulphate solution?

(A) Ag^+ (B) Zn^{2+}

(C) Pb^{2+} (D) Bi^{3+}

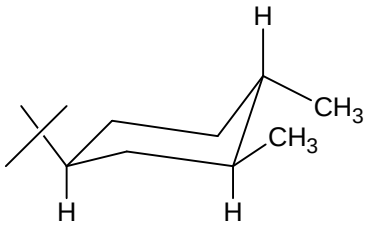
10. Select the reaction, which is not disproportionation reaction?

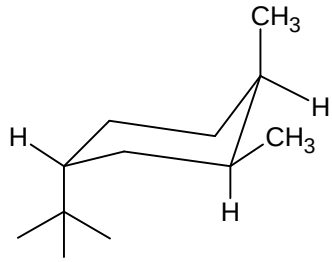
(A) $NH_4NO_3 \xrightarrow{\Delta} ?$ (B) $Na_2SO_3 \xrightarrow{\Delta} ?$

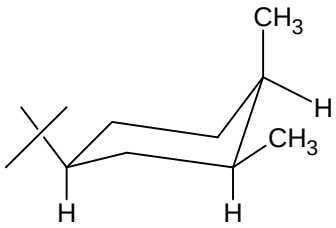
(C) $P_4 + NaOH \xrightarrow{\Delta} ?$ (D) $H_3PO_3 \xrightarrow{\Delta} ?$

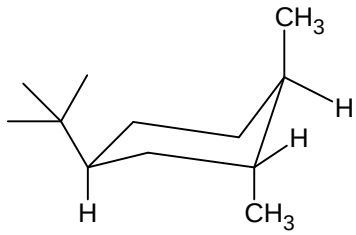
Rough Work

11. Which of the following conformations has maximum energy?

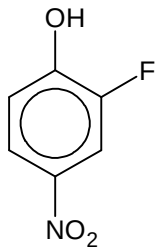
(A) 

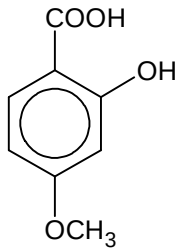
(B) 

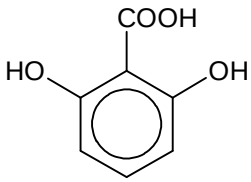
(C) 

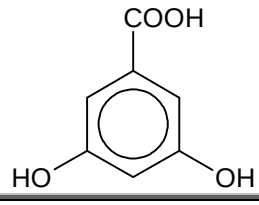
(D) 

12. The most acidic compound among the following is:

(A) 

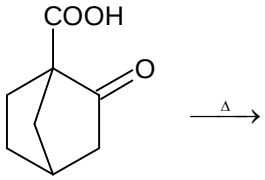
(B) 

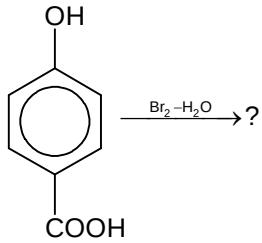
(C) 

(D) 

Rough Work

13. Which of the following reactions would produce a gas which turns lime water milky?

(A) 

(B) 

(C) $MgCl_2.6H_2O \xrightarrow{\Delta}$

(D) All of these

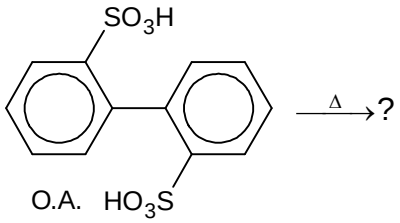
14. $CF_3I + NaOH(\text{aqueous}) \longrightarrow ?$

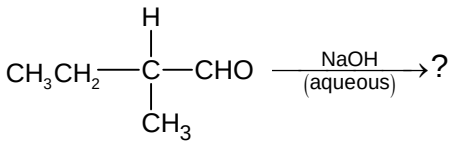
The major product formed is:

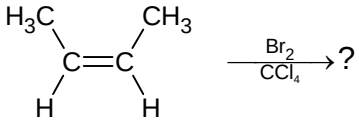
(A) CF_3OH (B) $F_2C = CF_2$

(C) CHF_3 (D) $F_3C - OCF_3$

15. The reaction (s) which would yield racemic mixture is/are:

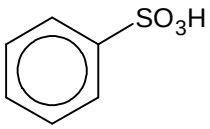
(A) 

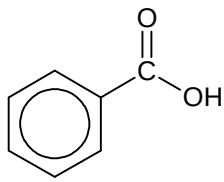
(B) 

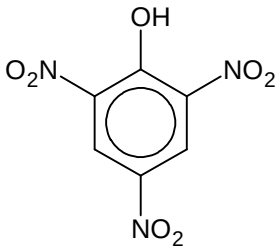
(C) 

(D) All of these

16. Which of the following would produce effervescence with sodium bicarbonate?

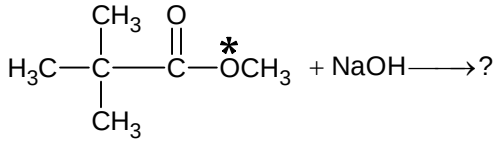
(A) 

(B) 

(C) 

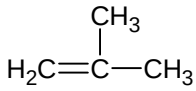
(D) All of these

Rough Work

17. 

The major product formed in the above reaction is:

(A) $CH_3\overset{*}{O}H$ (B) CH_3OH

(C) $(CH_3)_3C - COONa$ (D) 

18. PVC is the polymer of

(A) $CH_2 = CH - CH_2Cl$ (B) $CH_2 = \underset{Cl}{C} - CH_3$

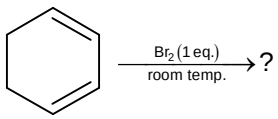
(C) $CH_2 = CH - Cl$ (D) $F_2C = CF_2$

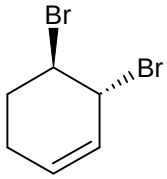
19. Mutarotation is not exhibited by:

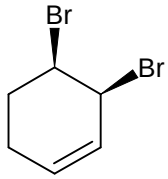
(A) α-D-glucopyranose (B) Methyl α-D-glucopyranoside

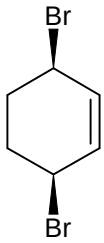
(C) α-D-fructofuranose (D) All of these

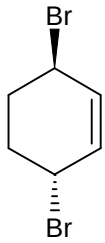
20. The product which is least expected to be produced in significant yield in the reaction given below, is:



(A) 

(B) 

(C) 

(D) 

21. A monoatomic ideal gas expands by doing q/2 work as a result of absorption of 'q' amount of heat. The molar heat capacity for the process is:

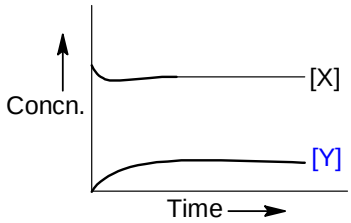
(A) R (B) 2R

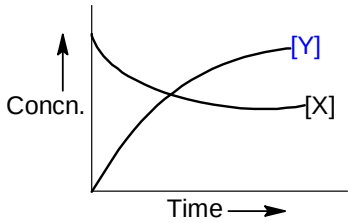
(C) 3R (D) 4R

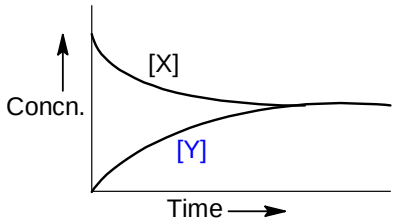
Rough Work

22. A metallic oxide contains 20% oxygen by mass. The equivalent weight of metal is:
(A) 40 (B) 32
(C) 16 (D) 48

23. The correct representation for the reaction $X \rightleftharpoons Y$, starting with X would be:

(A)


(B)


(C)


(D) Any of these

24. pH of aqueous solution of one of the following salts is independent of its concentration. The salt is:
(A) CH_3COONa (B) $C_6H_5NH_3Cl$
(C) $PbCl_2$ (D) CH_3COONH_4

25. The total volume of gases liberated at S.T.P. when a charge of 2F is passed through aqueous solution of sodium phosphate using inert electrode is:
(A) 44.8 L (B) 22.4 L
(C) 33.6 L (D) 11.2 L

26. Which of the following orbital has equal probability of finding an electron in any direction at a certain distance from the nucleus?
(A) s-orbital (B) p-orbital
(C) d-orbital (D) f-orbital

Rough Work

27. Which of the following reactions is not feasible?

(A) $Cu + H_2SO_4\,(conc.) \longrightarrow CuSO_4 + SO_2 + 2H_2O$

(B) $Cu + H_2SO_4\,(dilute) \longrightarrow CuSO_4 + H_2$

(C) $Mn + HNO_3\,(very\ dilute) \longrightarrow Mn(NO_3)_2 + H_2$

(D) $Al + Mn_3O_4 \longrightarrow Mn(NO_3)_2 + H_2$

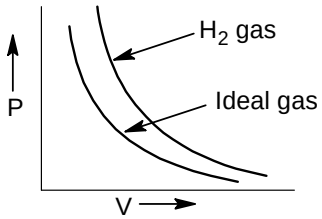
28. The total number of third nearest neighbours of Na^+ in NaCl lattice is:

(A) $6Cl^-$ (B) $12Na^+$

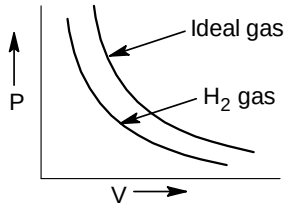
(C) $8Cl^-$ (D) $24Cl^-$

29. The correct graph for H_2 gas is:

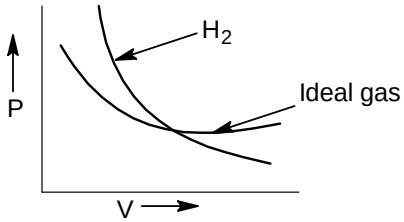
(A)



(B)



(C)



(D) All are correct

30. Which of the following binary liquid solutions would exhibit negative deviation from Raoult's law?

(A) Acetone + chloroform (B) Benzene + toluene

(C) Diethyl ether + acetone (D) Ethanol + acetone

Rough Work

Mathematics PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The function $f(x) = \frac{x^2 - 3x + 2}{x^2 + 2x - 3}$
   (A) has a maximum value at x = –3 (B) has a minimum value at x = 3
   (C) is increasing in its domain (D) is decreasing in its domain

2. The total number of 4 digit numbers which have digits in decreasing order (from left to right) such that number is divisible by 4 is
   (A) 110 (B) 118
   (C) 112 (D) 111

3. If in a triangle $R(b + c) = a \cdot \sqrt{bc}$, then the triangle is
   (A) equilateral (B) obtuse angled
   (C) right angled (D) nothing can be said

4. A tangent is drawn to a parabola at the point (12, 6) to meet the directrix at $\left(2, \frac{8}{3}\right)$. If focus lies on x–axis, then equation of parabola is
   (A) $y^2 = 4x$ (B) $y^2 = 4(x - 3)$
   (C) $y^2 = 4(x + 3)$ (D) none of these

5. Total number of triangles formed by the curves $x^3 - x^2 - x - 2 = 0$ and $4x^2 + x^2y^2 + 2y^2 - 3xy^2 - 12x + 8 = 0$ is equal to
   (A) 1 (B) 0
   (C) 2 (D) 3

Space for rough work

6. For a acute angle triangle, the value of $\frac{\sin A}{A}+\frac{\sin B}{B}+\frac{\sin C}{C}$ is

(A) greater than $\frac{6}{\pi}$ (B) $\frac{6}{\pi}$

(C) less than $\frac{6}{\pi}$ (D) none of these

7. An ellipse has the point (1, –1) and (2, –1) as its focii and x + y = 5 as one of its tangents, then the coordinates of the point where this line touches the ellipse is

(A) $\left(\frac{34}{9}, \frac{11}{9}\right)$ (B) $\left(\frac{32}{9}, \frac{13}{9}\right)$

(C) $\left(\frac{34}{9}, \frac{79}{9}\right)$ (D) $\left(\frac{32}{9}, \frac{77}{9}\right)$

8. In the process of numbering the pages of a large book, the printer uses 1890 digits. The number of pages in the book is

(A) 600 (B) 666

(C) 659 (D) 660

9. If the middle term in the expansion of $(1 + x)^{2n}$ is the greatest term, then the value of [x] can be [.]

(A) {0, 2} (B) {1, 2}

(C) {0, 1} (D) {0, 1, 2}

10. The value of $\sum_{n=1}^{2008}\frac{\sqrt{n^4+2n^3+3n^2+2n+1}}{n(n+1)}$ is equal to

(A) $2008+\frac{2008}{2009}$ (B) $2007+\frac{2008}{2009}$

(C) $2008+\frac{2007}{2009}$ (D) $2008+\frac{2007}{2008}$

11. Number of solutions of the equation $\sin^{-1} x + \cos^{-1}(x^2) = \frac{\pi}{2}$ is/are

(A) 0 (B) 1

(C) 2 (D) infinite

Space for rough work

12. The range of $y = \sin^3 x - 6\sin^2 x + 11\sin x - 6$ is
(A) [–24, 2] (B) [–24, 0]
(C) [0, 24] (D) none of these

13. If p and q are prime numbers and the equation $x^2 - px + q = 0$ has distinct integral roots then which of the following statements are true
(I) the difference of roots is odd
(II) at least one root is prime
(III) $p^2 - q$ is prime
(IV) p + q is prime
(A) I only (B) II only
(C) II and III only (D) all are true

14. Let x_1 and x_2 are two solutions of the equation $e^x \cos x = 1$, then the minimum number of the solutions of the equation $e^x \sin x = 1$ lying in between x_1 and x_2 can be
(A) 0 (B) 1
(C) 3 (D) none of these

15. The number of integral solutions of the equation $4\int_0^{\infty} \frac{\ln x}{x^2 + t^2} dt - \pi \ln 2 = 0$; x > 0 is/are
(A) 2 (B) 0
(C) 1 (D) 3

16. The complex number z which satisfy the equations |z| = 1 and $\left|\frac{z - \sqrt{2}(1+i)}{z}\right| = 1$ is
(A) z = 1 (B) z = (1 + i)
(C) $z = \frac{1+i}{\sqrt{2}}$ (D) $z = -\frac{(1+i)}{\sqrt{2}}$

17. The two adjacent sides AB and BC of a cyclic quadrilateral ABCD are 2 and 5 units respectively and the angle between them is 60º. If the area of quadrilateral is $4\sqrt{3}$ sq. units, then area of the circle circumscribing the quadrilateral ABCD is
(A) 48 π (B) $\frac{19\pi}{3}$
(C) 12 π (D) $\frac{19\pi}{12}$

Space for rough work

18. Let A = (2, 3, 5), B = (–4, 3, –1) and C = (3, 5, –2), then the equation of plane perpendicular to the plane ABC and passes through A, B is
(A) $2x - y + 2z + 13 = 0$ (B) $x + 2y - 2z + 3 = 0$
(C) $2x + y - 2z + 3 = 0$ (D) $2x - y + 2z - 11 = 0$

19. If A and B are two non singular matrices which commute then $(A(A + B)^{-1}B)^{-1}(AB)$ is equal to
(A) $A + B$ (B) $A^{-1} + B^{-1}$
(C) $A^{-1} + B$ (D) $A + B^{-1}$

20. Let P(x) be a polynomial of degree at most 5 which leaves remainders –1 and 1 upon division by $(x - 1)^3$ and $(x + 1)^3$ respectively then the number of real roots of $P(x) = 0$
(A) 1 (B) 3
(C) 5 (D) 2

21. If the equation $2x^2 + 2hxy + 6y^2 - 4x + 5y - 6 = 0$ represents a pair of straight lines, then the length of intercept on the x–axis cut by the lines is equal to
(A) 2 (B) 4
(C) $\sqrt{7}$ (D) 0

22. A polynomial is said to be a cyclotomic polynomial of degree $n > 1$ if it is of the form $x^n + x^{n-1} + \ldots + x + 1$. If $r_1, r_2, \ldots r_{100}$ are the roots of a cyclotomic polynomial of degree 100 then $(1 - r_1)(1 - r_2) \ldots (1 - r_{100})$ is equal to
(A) 100 (B) 101
(C) I (D) 0

23. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$, then $\vec{b}$ is equal to
(A) $2\hat{i}$ (B) $\hat{i} - \hat{j} + \hat{k}$
(C) $\hat{i}$ (D) $2\hat{j} - \hat{k}$

24. 5 different games are to be distributed among 4 children randomly. The probability that each child gets at least one game is
(A) $\frac{1}{4}$ (B) $\frac{15}{64}$
(C) $\frac{21}{64}$ (D) none of these

Space for rough work

25. If $\int_{-\pi/4}^{3\pi/4} \frac{e^{\pi/4}dx}{(e^x + e^{\pi/4})(\sin x + \cos x)} = k\int_{-\pi/2}^{\pi/2} \sec x dx$, then the value of k is

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

26. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{\lambda}$ and $\frac{x-1}{\lambda} = \frac{y-4}{2} = \frac{z-5}{1}$ intersect at a point (α, β, γ), then

(A) $\lambda = -1$ (B) $\lambda = 2$

(C) $\lambda = -3$ (D) $\lambda = 1$

27. The value of $\int_{-3}^{-1} (x+1)^4 (x+2)^3 (x+3)^4 dx$ is

(A) $\frac{7}{4}$ (B) $-\frac{3}{4}$

(C) 0 (D) none of these

28. The number of ways in which 9 identical balls can be placed in three identical boxes is

(A) 18 (B) 15

(C) 19 (D) 12

29. If $b_{n+1} = \frac{1}{1-b_n}$ $\forall$ $n \geq 1$ and $b_1 = b_3$ then $\sum_{r=1}^{2001} b_r^{2001}$ is equal to

(A) 2001 (B) –2001

(C) 0 (D) none of these

30. The value of $\int_{-1}^{0} \frac{x^2 + 2x}{\ln(x+1)} dx$ is equal to

(A) ln 2 (B) ln 4

(C) ln 3 (D) ln 6

Space for rough work